

Exploratory Data Analysis (EDA)

Salary Dataset

Tools: Python, Pandas, Matplotlib, Seaborn, SciPy

Data: Cleaned salary dataset from Task 1 (1,783 rows; Age, Gender, Education Level, Job Title, Years of Experience, Salary)

1. Objective

Explore the salary dataset using statistical summaries and visualizations to uncover patterns, correlations and the key factors that influence salary.

2. Statistical Summary

Variable	Mean	Median	Std Dev	Min	Max	Skew
Age	35.2	33	8.2	21	62	0.67
Years of Experience	9.2	8	6.8	0	34	0.90
Salary (\$)	113,399	110,000	51,361	25,000	250,000	0.09

Age and Experience are right-skewed (more younger / less-experienced employees, with a long tail toward senior staff). Salary is almost symmetric (skew close to 0), so the mean and median are close together (\$113K vs \$110K).

Gender: Male 964, Female 812, Other 7. **Education:** Bachelor's 766, Master's 568, PhD 339, High School 110.

3. Correlation Analysis

	Age	Years of Experience	Salary
Age	1.00	0.94	0.77
Years of Experience	0.94	1.00	0.82
Salary	0.77	0.82	1.00

Years of Experience has the strongest correlation with Salary ($r = 0.82$), followed closely by Age ($r = 0.77$). Age and Experience are themselves highly correlated ($r = 0.94$), so they carry largely the same information.

4. Statistical Significance Tests

- **Gender vs Salary (independent t-test):** $t = 4.39$, $p < 0.0001$ → the difference between average male and female salary is statistically significant, though the gap is modest (about 10%) and does not account for role or experience.
- **Education vs Salary (one-way ANOVA):** $F = 357.4$, $p < 0.0001$ → salary differs significantly across education levels; this is the strongest categorical driver found in the data.

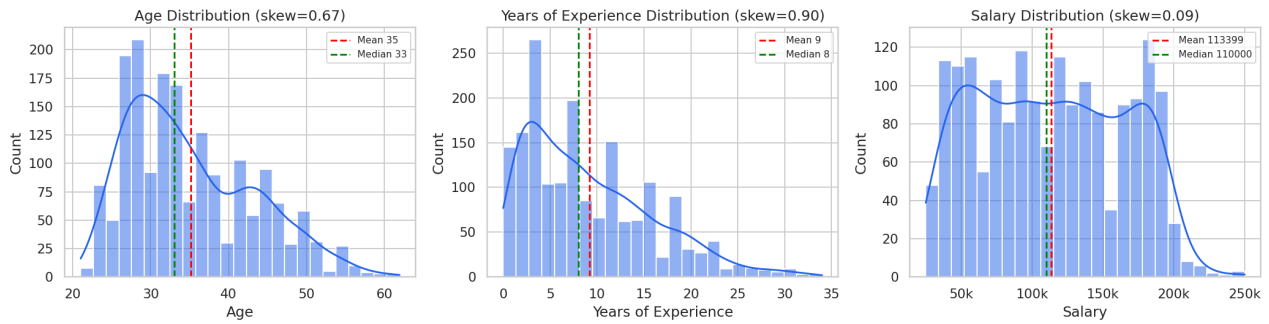
5. Key Influencing Factors

1. **Years of Experience** is the single biggest driver of salary ($r = 0.82$); pay rises steadily and then plateaus after about 20 years.
2. **Education level** creates large, statistically significant pay gaps: High School (~\$45K) to PhD (~\$161K).
3. **Job role / seniority** matters: director-level titles average \$170K-200K, well above individual-contributor roles.

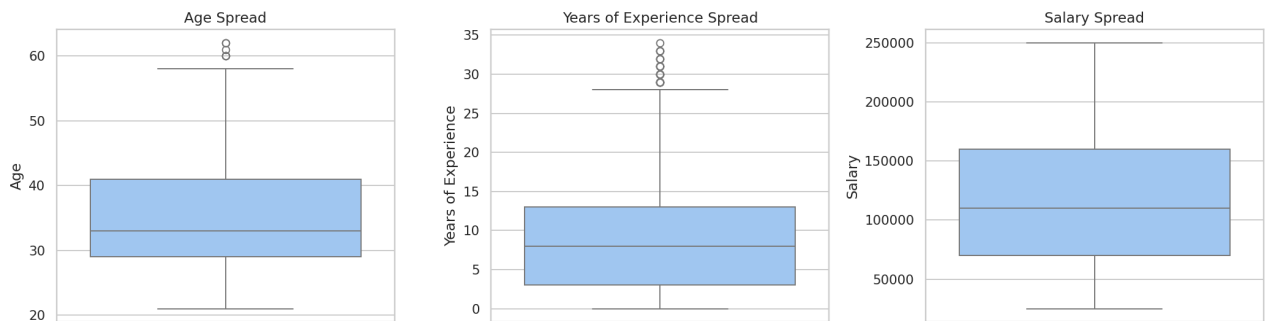
4. **Gender** shows a statistically significant but comparatively small gap; it is not a major driver next to experience and education.

6. Visualizations

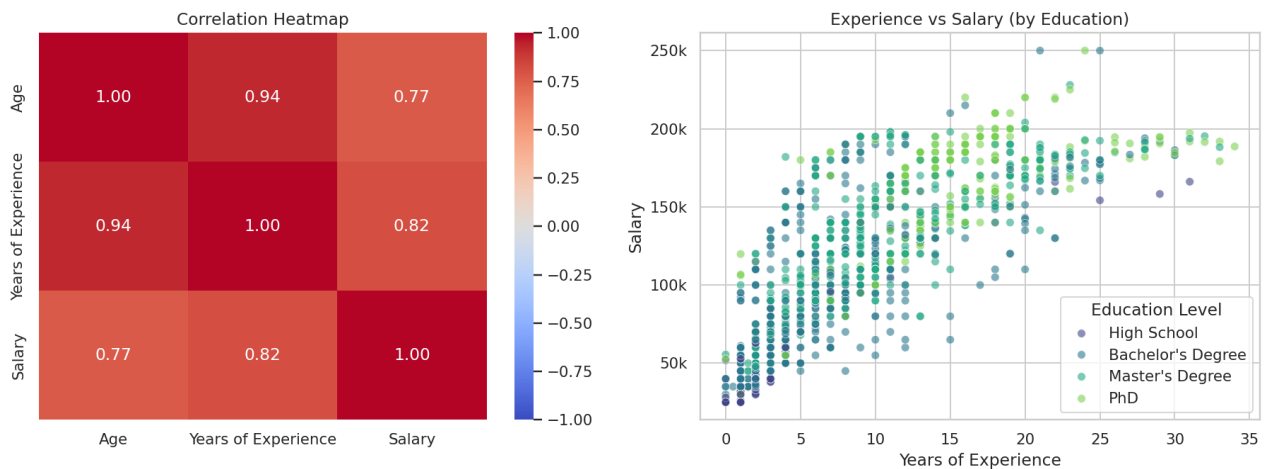
Distributions of Age, Experience and Salary



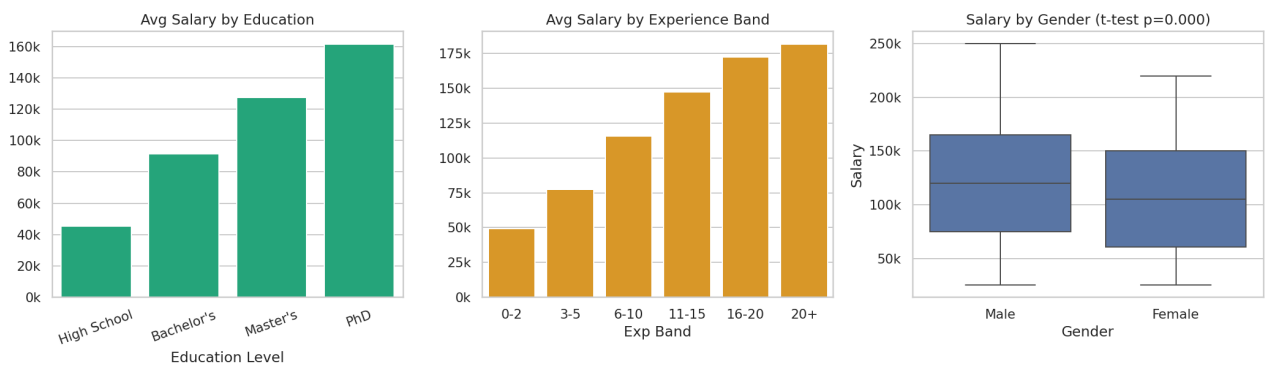
Spread and Outlier Check



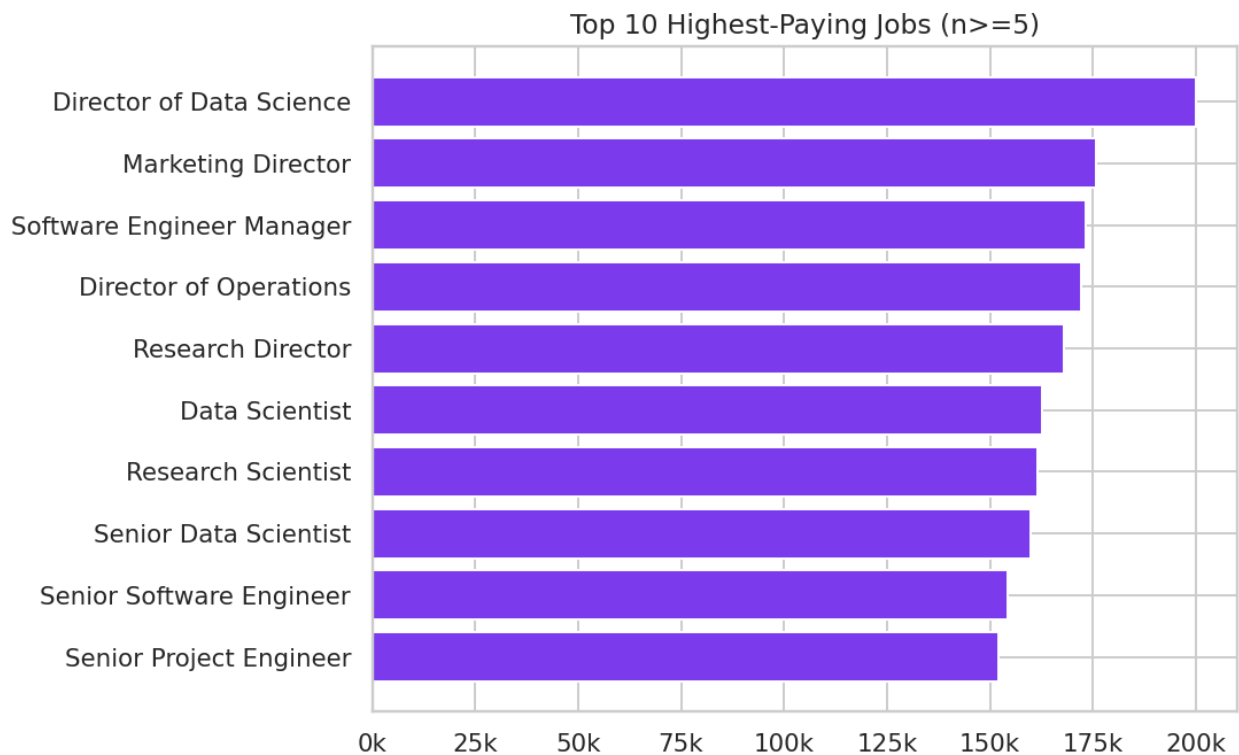
Correlation Heatmap and Experience vs Salary



Salary by Education, Experience Band and Gender



Top 10 Highest-Paying Job Titles



7. Conclusion

The exploration shows Salary is driven mainly by Years of Experience and Education Level, with Job Title/seniority also playing a major role. Age adds little beyond what Experience already explains, since the two are highly correlated. Gender shows a small but statistically significant gap that is worth monitoring but is not a primary driver of pay in this dataset.